

10-15

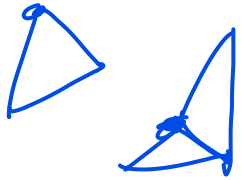
Lec_64

5.MST (minimum spanning tree)

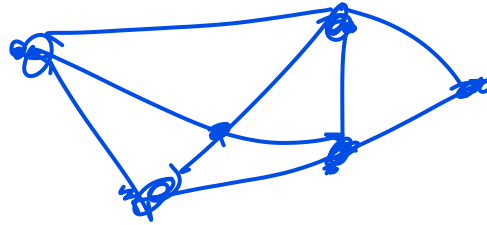
40-15

Let's see if the greedy

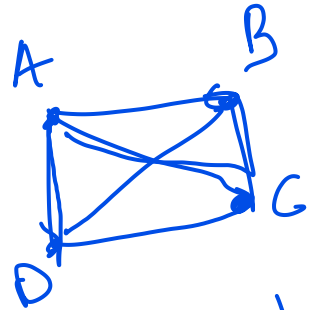
What is Graph?: graph is represent thing nodes and



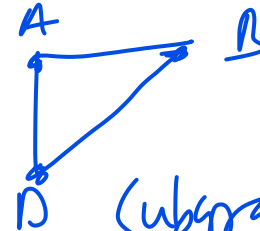
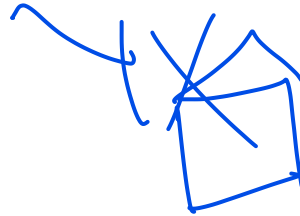
edges



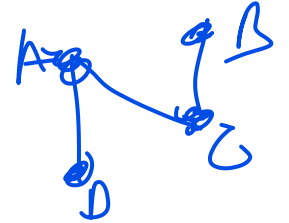
• SubGraph:



graph



subgraph



Spanning tree:

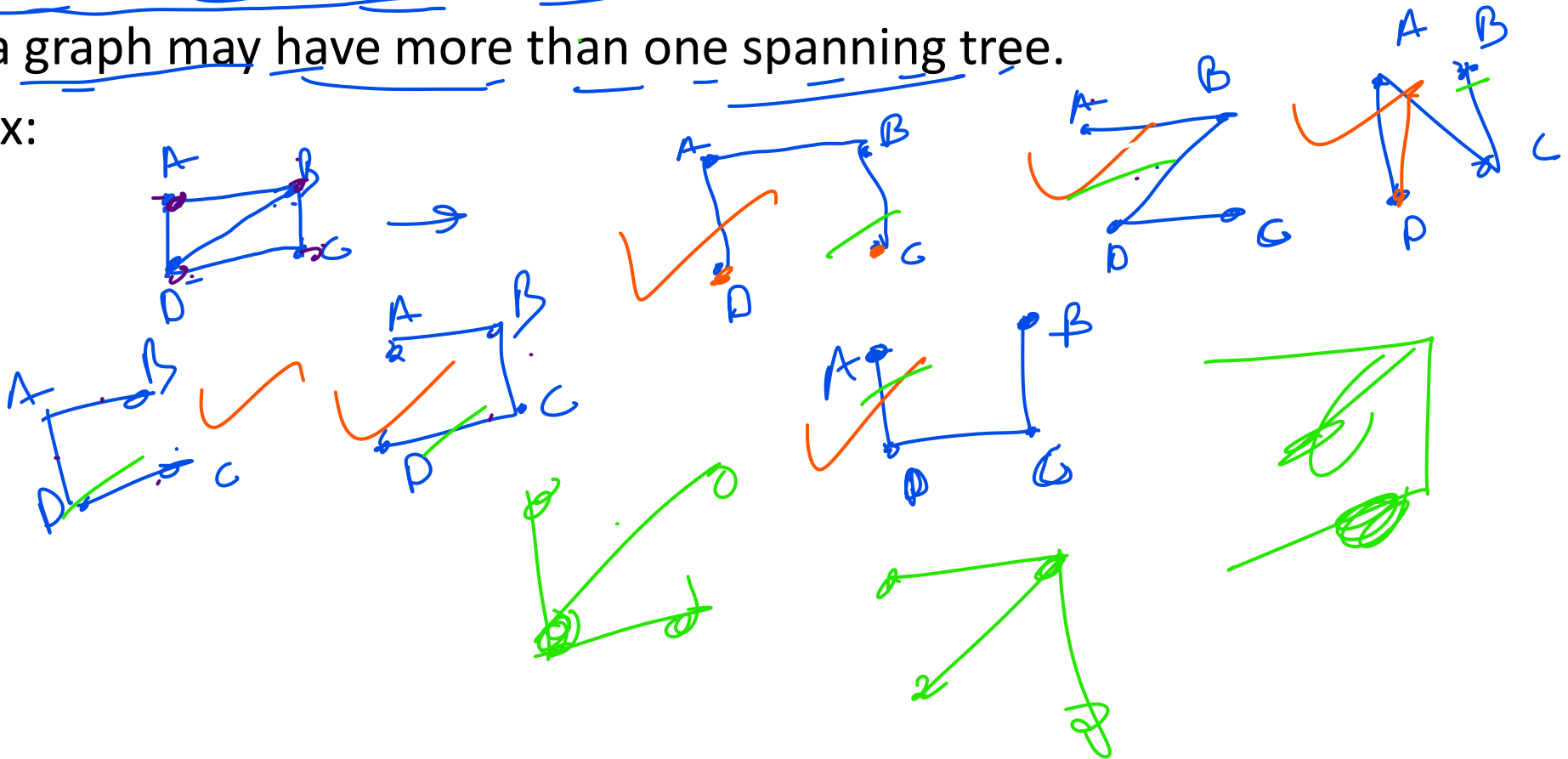
all vertices

subgraph

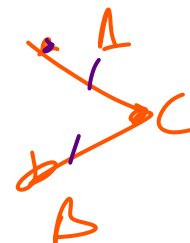
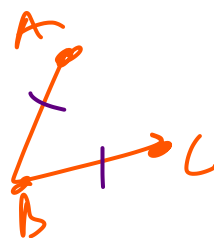
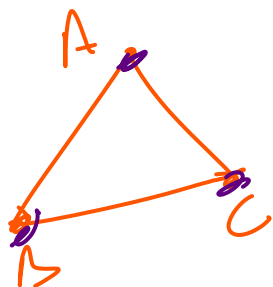
should be
reachable to
each other

(nsf)

- A spanning tree is a subset of Graph G, such that all the vertices are connected using minimum possible number of edges. Hence, a spanning tree does not have cycles.
- a graph may have more than one spanning tree.
- Ex:



Examples:



Complete graph:-

all vertices should be connected to each other

Degree =
a property of vertex

K_1

K_2

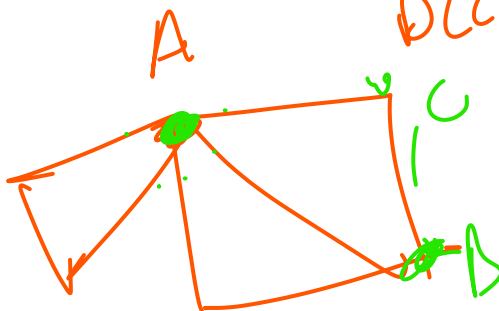
K_3

K_4

K_5

$D(A) = 2$
 $D(B) = 2$
 $D(C) = 2$

$D(C) = 2$



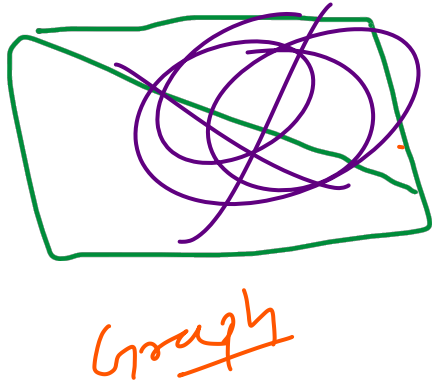
$D(A) = 5$
 $D(B) = 3$

In the spanning tree of a graph having vertices V ~~total~~ V vertices and $V-1$ edges

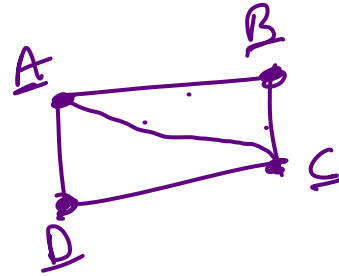
Counting number of Spanning trees

• 1.)

$$= \frac{n!}{n-1 \cdot 1!}$$



no of vertices = 4
no of edges = 5



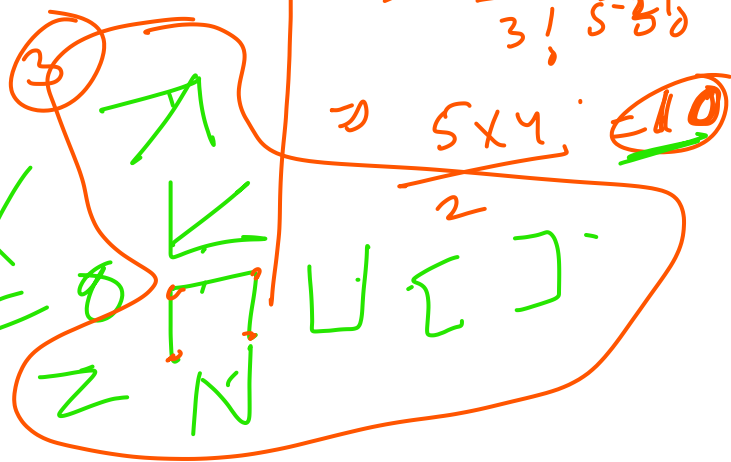
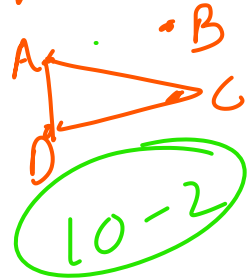
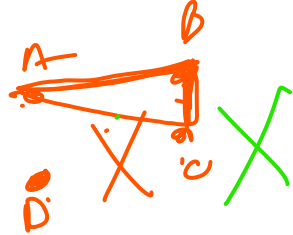
vertices = 4
now many edges should be
in the ST = $4 - 1 = 3$

out of 5 edges we
need to choose 3
edges

$$\Rightarrow \frac{5!}{3! \cdot 2!} = \frac{5!}{3! \cdot 2!}$$

Spanning tree $\mathcal{L}$ vertex = 4
edges = $4 - 1 = 3$

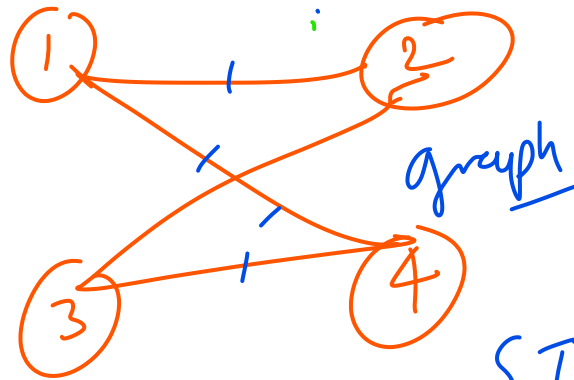
cycle checking
①



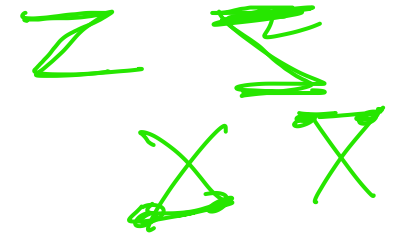
In a graph if n vertices and $n-1$ edges are there
if cycle is there, then graph is not connected
else graph is connected

Count How many spanning trees.

2.) ∴



$$\begin{cases} V = 4 \\ E = 4 \end{cases}$$



$$\begin{cases} V = 4 \\ E = 4 - 1 = 3 \end{cases}$$

$$4(3) \rightarrow$$

$$\frac{4!}{3!1!}$$

$$=$$

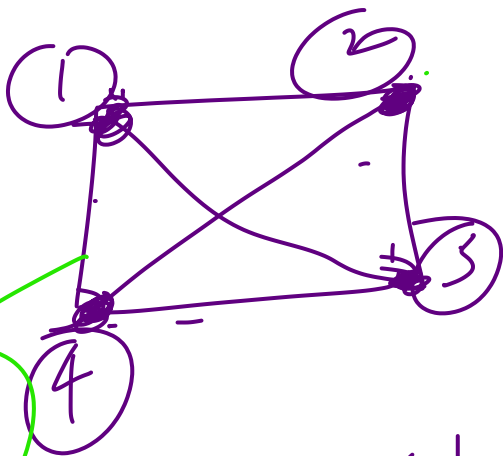
$$\frac{4 \times 3!}{3!1!}$$

$$=$$

$$4$$

3.)

complete graph



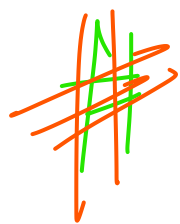
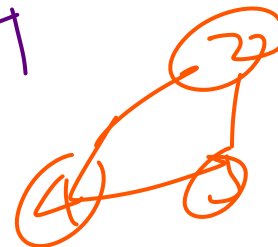
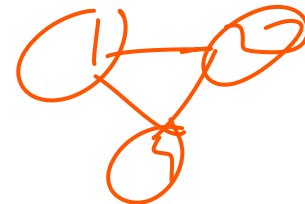
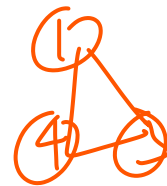
graph $\begin{cases} V = 4 \\ E = 6 \end{cases}$

ST $\begin{cases} V = 4 \\ E = 3 \end{cases}$

$$6C3 = \frac{6!}{3!3!} =$$

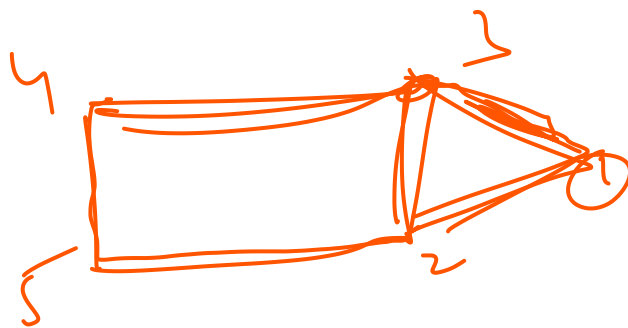
$$20 - 4 = 16$$

$$\frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$$



In a complete graph of n vertices n
no. of ST = $n^{n-2} \Rightarrow 4^{4-2} = 4^2 = 16$

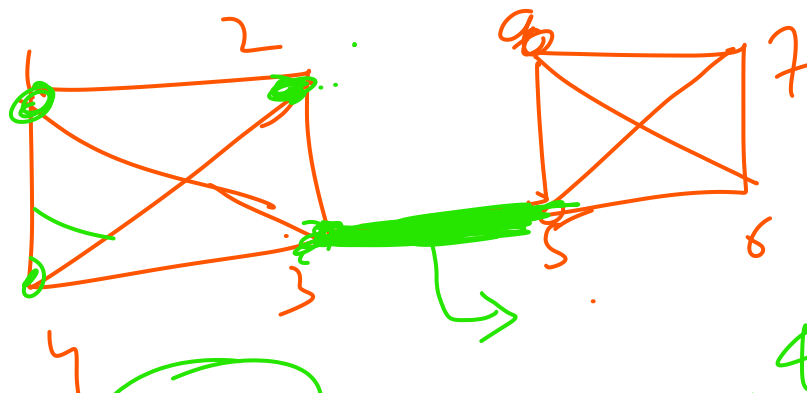
4.)



How many ST 2

(cut)

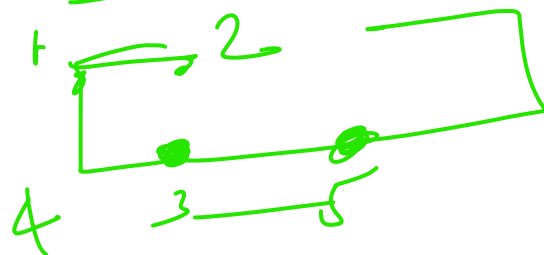
5.)



how many ST?

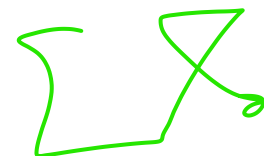
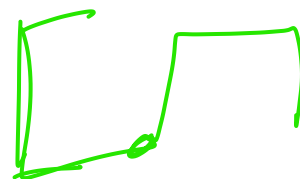
$$\begin{array}{c} 4-2 \\ 4 \end{array}$$

$$4^2 \times 4^2 =$$



$$4^{4-2}$$

$$4^4 = 16 \times 16 = 256$$



min Spu Tree

